

Accessibility-Related Publication Distribution in HCI Based on a Meta-Analysis

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ABSTRACT

Accessibility research aims to aid humans that experience minor or major disabilities and conditions. However, researchers might have limited exposure to certain disabilities, therefore, focus on those prevalent in their own lives. This work presents a script-based meta-analysis on addressed populations in accessibility research published on top Human-Computer Interaction (HCI) venues (3617 full papers). We categorize the publications regarding the involved people and their disabilities. We found that work on *vision disability* makes up for almost one third (27.85%) of the work published in general HCI. In light of these findings, we present possible conference- and funding-related explanatory approaches and argue that disability research could more reflect the prevalence of disabilities in the world.

CCS CONCEPTS

• General and reference → Surveys and overviews.

KEYWORDS

Accessibility; overview; survey.

ACM Reference Format:

Mark Colley, Taras Kränzle, and Enrico Rukzio. 2022. Accessibility-Related Publication Distribution in HCI Based on a Meta-Analysis. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29-May 5, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 28 pages. <https://doi.org/10.1145/3491101.3519701>

1 INTRODUCTION

According to *The International Classification of Functioning, Disability and Health* (ICF) [316], there are multiple categories of impairments that can be broadly sub-categorized into four types [129]: Mobility and Physical Impairments, Vision, Hearing, and Cognitive or Learning Disabilities. For example, there are around 1.3 billion people with some, 217 million people with a severe vision impairment, and 36 million people are blind [42, 247]; 466 million people have disabling hearing loss [248], and 50 million people have some form of dementia [249]. The distribution of disability in the USA in 2015, as an example, is 3.6% for hearing, 2.3% for vision, and 4.8%

for cognitive impairments [176]. Accessibility plays an important role in various conferences today both for attendance as well as a research topic [1]. The *International ACM SIGACCESS Conference on Computers and Accessibility* (ASSETS) [90] specifically addresses research on people with disabilities. Additionally, there are special subcommittees at CHI for mental health, learning and accessibility & ageing [245]. However, a quantitative approach to how strongly each disability is addressed in the field of Human-Computer Interaction (HCI) is missing. HCI refers to a “multidisciplinary field of study focusing on the design of computer technology and, in particular, the interaction between humans (the users) and computers” [98]. This is important to shape future research. Complex problems require formulating the challenges first [134]. Knowing the distribution of papers on disabilities can facilitate discussions about potential underrepresentation, especially important research topics, and distribution of research funds. Thus, the first important step is to report relevant data as, for example, done by Microsoft [185].

This work first provides an overview of the addressed population from 2009 – 2019 that were published at ASSETS to define relevant fields and terminology. Based on this overview, a script-based categorization approach was defined. For this, we first categorized the papers of ASSETS manually and then compared this to the script-based paper categorization approach. Such a script-based approach enabled us to categorize many more publications than would be possible manually. After an iterative improvement process, we used this script to search further venues in the accessibility community (W4A [6], TACCESS [243]) and the broader HCI domain (CHI [244], UIST [293], AutoUI [149], and MUM [216]) in the time frame of 2016 – 2019 were included in the quantitative categorization process. We found that conferences vary greatly in the addressed populations such as visual disabilities for ASSETS, TACCESS, and W4A or cognitive and learning disabilities at CHI. Visual impairments were studied and addressed the most (27.85%).

Contribution statement: This work contributes to the body of accessibility-related knowledge by providing a meta-analysis on addressed disabilities in the field of HCI. For this, an R script-based approach was defined, compared to a manual classification of the ASSETS papers from 2009 – 2019 and then applied to other relevant venues. This work shows different focus points of the conferences and reveals the misalignment of the work on disabilities with the prevalence of disabilities in the general population.

2 METHOD

In this paper, we provide a meta-analysis on the addressed disabilities at major HCI venues. First, we describe our manual and script-based process. We reviewed the main proceedings of ASSETS



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CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA
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ACM ISBN 978-1-4503-9156-6/22/04.
<https://doi.org/10.1145/3491101.3519701>

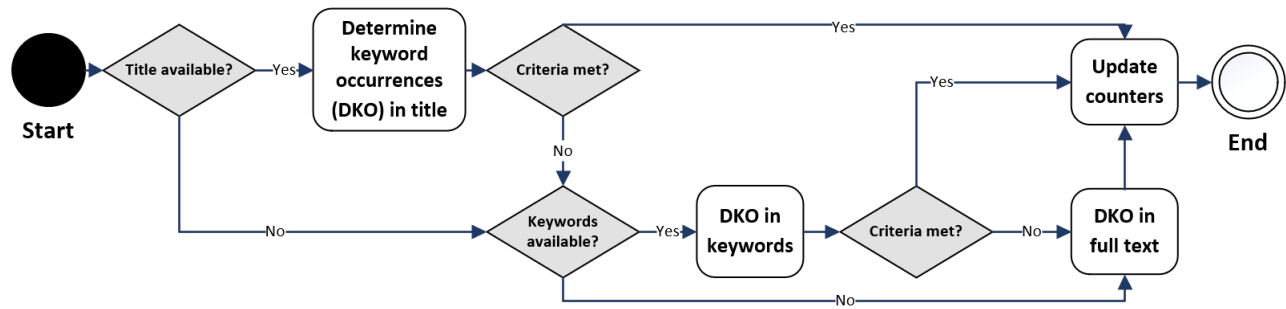


Figure 1: Classification procedure in the R script.

from 2009 – 2019 (316 publications). We collected all 316 ASSETS papers and formalized our process to classify and review the papers. We chose ASSETS as a premier venue for accessibility-related publications and the time frame to gather a large enough sample.

2.1 Manual Categorization

All ASSETS papers were categorized manually by two authors. For this, the two authors first read the title, the abstract, and the keywords. If a categorization was not possible based on these sections, the entire full paper was read. Both authors categorized all papers. Disagreements were resolved via discussions. One author already has knowledge about and worked with people with vision impairments. Additionally, a medical physician was available for queries regarding nomenclature and disabilities. First, we categorized publications into four types [129]: Mobility and Physical Impairments, Vision, Hearing, and Cognitive or Learning Disabilities. We found that some papers could not be categorized as more than one disability or older adults were addressed. Therefore, we added the categories *General Accessibility* and *Elderly*. We explicitly do not see being of older age to be a disability. Additionally, we divided psychological disorders from cognitive disabilities as these seem to become more prevalent [246]. This is also in line with the distinction of the UN Convention on the Rights of Persons with Disabilities (UNCPRD) between mental and intellectual disabilities [121]. As we will discuss in section 5, there is an inevitable conceptual overlap between these disabilities [208], however, not in our sub-category terminology. The results can be seen in Table 1.

2.2 Script-based Categorization

After the manual categorization, the publications were analyzed via a R script using the *pdfsearch* [187] (see Appendix C; available under <https://github.com/M-Colley/accessibility-distribution-meta>). We defined relevant terms of the addressed disability (e.g., “visual impairment”). Additionally, we asked a medical physician for related keywords. As authors tend to use different vocabulary, we used the first ten synonyms with a rating greater than 200 (highest possible, crowd-sourced) from <https://www.powerthesaurus.org> as an “excellent resource of English words and their synonyms” [281, p. 4] for the search.

With preliminary criteria, we first analyzed the 316 publications of ASSETS. In an iterative process, we added relevant keywords.

When no match was found, the paper was again analyzed independently by two authors and relevant search terms were added to the script. Therefore, the search term lists vary in length. Disagreements were resolved through discussions. In the case of unclear affiliations to one of the categories, the authors also asked a medical physician as these are examined in detecting and working with people with disabilities.

For ASSETS, the classification based on the R script for ASSETS was then compared to our manual categorization. Our script categorized 306/316 ($\approx 96.84\%$) ASSETS paper. We found that 59/77 table cells (one cell per year and addressed disability; e.g., *Vision Disability* for 2009 had 9 references in Table 1) were populated the same. As a distance function measuring the deviation from the manual categorization, we defined the absolute difference for each cell. This showed that there is a difference of 22 (max 2 per cell, see Appendix B Table 3) publications (e.g., the automatically generated table has 7 references for *Vision Disability* for 2009, thus the difference is 2). Ten of these can be attributed to the non-classified publications and five can be attributed to non-searchable publications. Nevertheless, we were able to correctly classify 93% of all publications at ASSETS. This leads us to conclude that our approach, while not perfect, provides a reliable picture of accessibility-related publications at HCI venues. The relevant search terms can be found in Appendix A.

Afterward, we repeated this with the additional publications. For this, we followed the PRISMA [217] process. PRISMA refers to the *Preferred Reporting Items for Systematic Reviews and Meta-Analyses*. The papers had to be published within the time frame of 2016 – 2019. We chose this different time frame because (1) we found no changes over the distribution at ASSETS over the 11 years and (2) CHI allows bulk downloads for this time frame (as of January 2021). They also had to be published in a journal or on a conference with an HCI focus: W4A, TACCESS, CHI, UIST, AutoUI, and MUM. We downloaded all papers from the ACM DL.

First, we repeated the procedure with the papers of W4A and TACCESS, as it is clear that all publications address accessibility. Again, relevant search terms from titles were added manually. After another iteration, with our script, of the 494 publication in ASSETS, W4A, and TACCESS, we could **not** categorize 15 or 3.23% of the publications. Of these, 5 were deemed *maybe-accessibility-related* and 10 *accessibility-unrelated* (see criteria below).

For these 494 publications, on average, the category with the most occurrences had $M=88.41$ ($SD=69.07$) occurrences (e.g., terms from one category occurred ≈ 88 times). The first quartile was 41. The category with the second most occurrences had $M=14.02$ ($SD=17.00$). The first quartile was 4. Overall, publications had $M=112.99$ ($SD=79.53$) accessibility-related keywords, the first quartile being 59. Based on these numbers, we considered a publication as assigned to a category when a search term was found in the title or the keywords or otherwise if (1) more than 23 references (0.1 quantile of most common keyword category) to one category were found and (2) either the ratio to the second most common keywords was above 2.0 or no more than 14 references (mean of second most common keywords) to another category were found in the entire full paper. *References* refers to the number of appearances of one specific term (e.g., *dyslexia*) in the **entire** full paper. As all works include some reference to accessibility, we defined that if at least **20** references (0.05-quantile of the overall keyword occurrences for the accessibility-related venues) to search terms are found, this paper was categorized as *maybe accessibility-related*. With this threshold, only $10/494 \approx 2.02\%$ publications of the accessibility-related venues ASSETS, TACCESS, and W4A were wrongly categorized as *accessibility-unrelated*. 271 were categorized by title, 46 by keywords. This approach also guards against “accessibility-as-icing”, as singular sentences indicating a potential relevance to people with disabilities will not be categorized as relevant.

As we aimed to gain a broader picture of the HCI community, we included the full papers of the conferences CHI, UIST, AutoUI, and MUM in the R-based analysis. We chose CHI and UIST as large top tier HCI venues and AutoUI and MuM as surrogates for smaller, more focused conferences. With the updated criteria, we checked **all** full papers. Taken together, we categorized **3617** full papers. For every conference, a data table was automatically generated (see supplementary material). Following the PRISMA method [217], we only included the papers outlined above, excluded no records, and performed no qualitative synthesis

Usage of pdfsearch [187] in full text:
`keyword_search(pdf_file, keyword=keywords, path=TRUE, remove_hyphen=TRUE, surround_lines=1, ignore_case=TRUE, split_pdf=TRUE)` where: *keywords* are the search terms per disability, *remove_hyphen* is set to TRUE to combine hyphenated words, *ignore_case* is TRUE to disregard capitalization, and *split_pdf* is TRUE as this is “most useful with multicolumn pdf files” [187, p. 6].

3 FINDINGS

We report descriptive statistics for all papers and conference-related findings. We found that **1062** of the 3617 (29.36%) full papers to address a disability. Additionally, we found that 400 full papers were *potentially accessibility-related*. 577 of 1062 (54.33%) were classified by either the title (445 or 41.90%) or the keywords (132 or 12.43%).

ASSETS: We predominantly found vision-related papers (128/316 or 40.51%; see Figure 2a). While we approve of this effort and do not want this to cease, we are worried that other disabilities are not included sufficiently. Hearing and cognitive disabilities were addressed the second most. All other disabilities each represent $\approx 10\%$ or less of the papers.

AutoUI: AutoUI did not address disabilities to a great extent. With our script, we found that one publication addressed cognitive disability and two addressed older adults (2.2% of the 136 analyzed papers).

CHI: 515/2514 (20.49%) papers addressed disability. These works mainly focused on older adults (128, 5.09%), psychological (94, 3.74%), or cognitive (86, 3.42%) disabilities (see Figure 2b). This accounts for the more frequent occurrence of mental health problems [246]. Visual disabilities were the fourth most common disability addressed (84). Other disabilities were only addressed to a minor degree; hearing disability, as the fifth most common, was only addressed in 4.47% of all the accessibility-related papers.

MUM: We found that two papers addressed vision, four cognitive, one psychological, five mobility disability and five addressed older adults (in total, 11.49% of the 148 analyzed papers).

TACCESS: We categorized 70/70 papers. Vision-related papers were most dominant (31/70, 44.29%). Older adults, the second most common addressed population, were only addressed in 10/70 (14.29%) of the categorized papers.

UIST: 49/325 (15.08%) addressed disability. Mobility (21, 6.46%) and visual (15, 4.62%) were most dominantly addressed.

W4A: 102/108 (94.44%) papers were categorized. W4A, with addressing website accessibility, focuses mainly on general accessibility (43, 39.81%; this includes accessibility evaluations) and visual impairments (33, 30.56%).

In Table 2, the occurrences per disability addressed are summed up. Most work focuses on vision impairments while the other disabilities are addressed to an equal degree. Hearing-impairment related work is scarcer. When the distribution of disabilities is taken into account, we argue that this distribution could look different and, therefore, while these findings are not worrisome, action should be taken to better address underrepresented disabilities.

4 EXPLANATORY APPROACHES, IMPLICATIONS, AND RECOMMENDATIONS

Almost a third of the publications were focused on vision impairment and another third on the combination of cognitive and psychological disabilities. In this section, we propose explanations for the analysis’ findings.

4.1 Possible Conference-Related Explanations

Driving is a highly visual and often demanding task which is difficult for people with impairments [250]. This could be one reason for the lack of accessibility-related work on AutoUI. However, there are approaches, e.g., to aid people with vision impairments to steer [300]. We also believe that with the advance of autonomous driving, the target group of vehicle-related research will change towards work-related and accessibility-related research (e.g., see [77]).

ASSETS, TACCESS, and UIST seem to focus highly on visual impairments. One interpretation of the dominance of visual-impairment related papers might be that a lot of visual feedback is apparent in today’s society, e.g., billboards, television, and the worldwide web. We predominantly perceive information through a display in 2020, for example via smartwatch, smartphone, tablet, laptop, PC, or screen walls. Therefore, researchers could attribute a high relevance to these topics. With an increased amount of visual

Table 1: Distribution of publications based on the addressed population at ASSETS from 2009 – 2019.

Year	Vision Disability	Hearing Disability	Brain, Cognitive or Learning Disabilities	Psychological Disorders	Mobility and Physical Impairments	General Accessibility	Elderly	Overall Papers
2009	9[5, 13, 17, 97, 119, 164, 278, 319, 324]	3[70, 196, 211]	4[27, 71, 142, 194]	3[125, 198, 235]	1[200]	5[33, 49, 107, 303, 340]	1[323]	26
2010	9[72, 132, 172, 190, 223, 228, 257, 277, 311]	4[144, 180, 308, 326]	6[11, 136, 189, 226, 254, 335]	3[120, 230, 236]	0	5[34, 50, 91, 177, 202]	1[138]	28
2011	8[86, 93, 155, 171, 204, 242, 299, 338]	5[135, 145, 170, 309, 325]	2[186, 320]	4[38, 41, 106, 124]	1[255]	4[2, 35, 215, 334]	3[76, 148, 279]	27
2012	10[21, 60, 94, 110, 141, 150, 182, 273, 318, 342]	4[181, 183, 197, 220]	1[315]	2[163, 165]	0	3[218, 260, 269]	5[88, 232, 237, 313, 333]	25
2013	12[20, 26, 69, 166, 184, 231, 240, 262, 305, 330, 339, 346]	4[158, 174, 310, 321]	3[137, 253, 265]	1[213]	3[89, 99, 314]	2[112, 256]	3[83, 109, 292]	28
2014	14[23, 24, 59, 62, 63, 92, 116, 151, 207, 214, 221, 239, 300, 329]	2[179, 307]	1[267]	2[87, 209]	4[66, 133, 219, 229]	2[45, 105]	4[22, 64, 81, 84]	29
2015	14[8, 40, 53, 104, 108, 117, 130, 233, 263, 272, 296, 317, 345, 349]	4[143, 159, 178, 288]	5[58, 74, 103, 156, 266]	3[210, 225, 337]	1[65]	4[79, 82, 102, 312]	1[80]	32
2016	12[3, 7, 14, 95, 114, 160, 199, 264, 294, 302, 306, 332]	4[46, 51, 167, 259]	4[173, 222, 284, 347]	2[271, 328]	0	2[205, 286]	0	24
2017	14[12, 36, 37, 52, 56, 115, 140, 191, 227, 241, 280, 285, 301, 348]	3[30, 146, 161]	3[75, 201, 268]	5[44, 127, 131, 206, 331]	2[67, 224]	1[274]	0	28
2018	13[4, 9, 39, 55, 57, 123, 169, 261, 287, 297, 298, 343, 344]	5[10, 48, 96, 153, 282]	3[18, 19, 25]	1[283]	1[68]	3[29, 128, 275]	2[192, 193]	28
2019	20[31, 54, 73, 111, 118, 122, 139, 154, 188, 203, 234, 252, 258, 276, 291, 295, 304, 322, 336, 341]	7[15, 47, 61, 152, 157, 162, 175]	3[28, 195, 290]	1[270]	3[16, 251, 289]	1[168]	6[32, 85, 100, 101, 147, 327]	41
Total	135 (42.72%)	45 (14.24%)	35 (11.08%)	27 (8.54%)	16 (5.06%)	32 (10.13%)	26 (8.23%)	316

Table 2: Distribution of overall papers based on addressed population.

Vision Disability	Hearing Disability	Brain, Cognitive or Learning Disabilities	Psychological Disorders	Mobility and Physical Impairments	General Accessibility	Elderly	Combined
291 (27.85%)	81 (7.75%)	145 (13.88%)	131 (12.54%)	121 (11.58%)	103 (9.86%)	173 (16.56%)	1045

displays, there is the possibility that people with visual disabilities are not able to fully make use of the possibilities.

Another possible interpretation is that visual impairments are potentially more “tangible” to seeing people than, for example, learning disabilities. It is rather easy to simulate vision loss either by closing one’s eyes or using simulations [113]. Simulations for physical impairments also seem possible with some more effort. This is, however, difficult for learning disabilities. Visual tasks also allow for relatively easy assessment and evaluation. Designing valid studies that can show effects on learning outcomes are subject to numerous confounding variables: Experts agreed that “further research involving technically adequate measures to determine cognitive, neuropsychological, and academic response-to-intervention interrelationships was necessary” [126, p. 231]. Additionally, “SLD [specific learning disabilities] need individualized interventions based on specific learning needs” [126, p. 223].

For CHI, we found that mostly older adults and cognitive and psychological disabilities were addressed. This seems to be in contrast to the given explanation for ASSETS and TACCESS. Visual impairments ranked only fourth. However, CHI is the leading venue in the HCI community, according to the Google scholar ranking and, therefore, attracts cutting edge research. Additionally, there are numerous subcommittees including *Learning, Education, and Families, Health*, and *Accessibility and Ageing* [245]. This could be a reason for the higher submissions of cognitive- and psychological-addressing research. W4A addresses web accessibility and its evaluation. As numerous websites are highly visual, this is a problem for people with vision impairments. Therefore, the focus on this population is explainable.

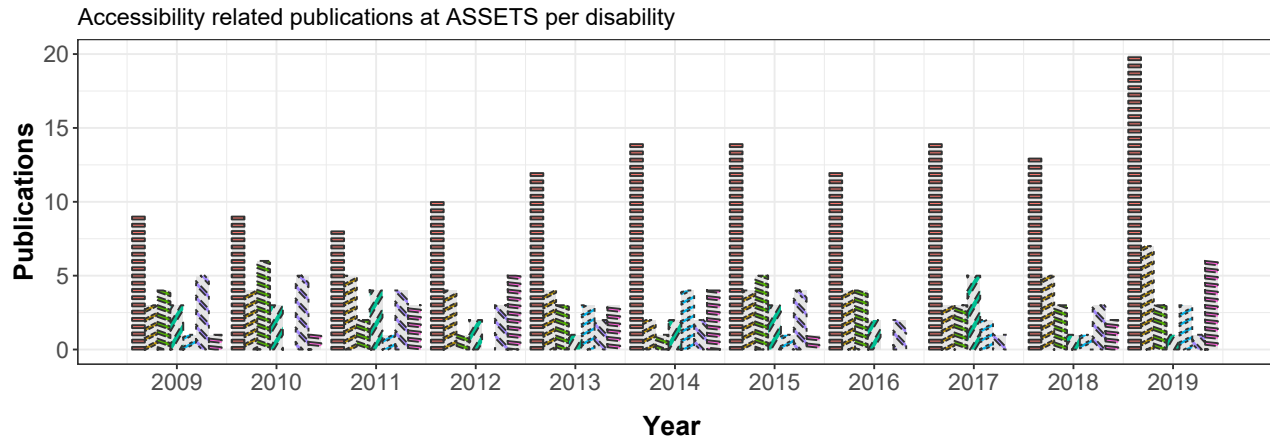
Finally, there are also difficulties regarding recruiting participants of certain disabilities and running usability testing, focus groups and interviews with younger/child populations.

4.2 Bias in Funding Agencies

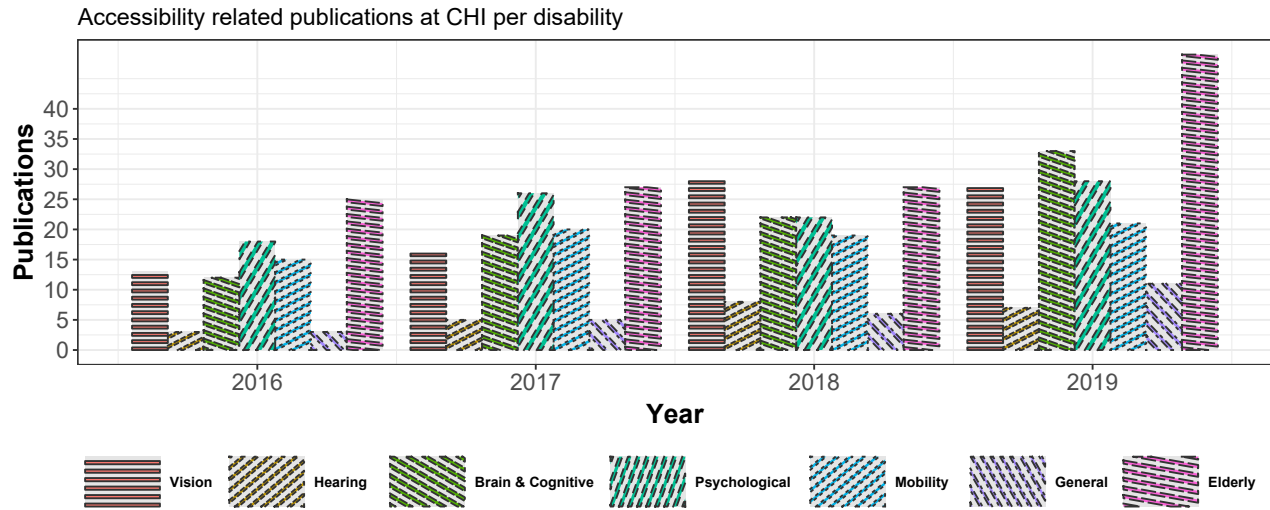
Funding of research projects is a potential source for the disproportionate distribution of papers. Currently, there are 25 EU-funded research projects on technologies for accessibility [78]. Three address visual impairments, two hearing impairments, one deaf-blindness, five motor impairments, nine cognitive impairments, and five general accessible communication technology. In Canada, currently announced projects are made of seven projects addressing motor impairment, eight projects addressing general disabilities, three visual, and two cognitive disabilities [238]. Microsoft’s *AI for Accessibility* program [212] has no special focus on a disability. In their featured examples, all categories of disabilities are addressed. While this is not a full view of funding agencies around the world, it seems that the bias towards visual and cognitive and psychological disabilities is, at least today, not induced by funding agencies.

4.3 Publication

Based on our script-based approach and the difficulties we experienced, we propose some enhancements for the publication process such as including the target group and the used methodology in the *Keywords*. This could also be done automatically, as done by other venues such as *IEEE Xplore Digital Library*. Additionally, we recommend a more rigorous approach to the enforcement of metadata validity. This is relevant for several tasks such as (our) script-based approaches but also, for example, bibliometrics. We also stress the



(a) Publications on ASSETS per disability from 2009 – 2019.



(b) Publications on CHI per disability from 2016 – 2019.

Figure 2: Publications on CHI and ASSETS per disability as a stacked bar chart.

need to provide papers in an accessible and searchable format. This is also necessary to ensure that screen readers can function properly.

5 THOUGHTS ON AND LIMITATIONS OF OUR APPROACH & FUTURE WORK

Categorization is shaped by culture and worldview and, itself shapes these [43]. It can also “suggest bias or reflect negative, disparaging, or patronizing attitudes toward individuals or groups of individuals” [129, p. 62]. With our suggested approach, potentially harmful considered terms (e.g., elderly, see [129]) were used for classification. Additionally, including a medical physician could have included a biased view on the topic of disability. Hanson et al. [129] state that

language alters and we strongly support this change towards un-harmful terminology, however, for our purpose, these search terms were necessary. Also, some overlap between the categorization is inevitable. We assured that most search terms can only be put into one category, however, for example, vision disability is often addressed for older adults, resulting in an overlap. The creation of the search terms, in general, was difficult and demanded an extensive use of reading at least titles and keywords of the paper. Still, it cannot be eliminated that some keywords were overlooked.

Our R-based categorization of work potentially could have mis-categorized some works due to unaccounted or overlapping search terms, not accounting for abbreviations etc. Additionally, the difference in length for search terms per disability could have biased the results. While the categorization via title or keywords, performed

well, it had difficulties when general terms were used in the keywords alongside more specific ones. For example, keywords that **first** have *assistive technology* and **then** *autism* would be categorized under *General Accessibility* while, if the keywords would be reversed it would be categorized under *Psychological Disorder*. This became clear when the script-based approach was compared to the manual categorization for ASSETS papers.

Classification solely based on the title, abstract, and keywords was difficult. While our script searched the entire paper, adding such relevant information in these parts would significantly ease the reading process. Also, we found that numerous files have incorrect metadata. This limited script-based process. For example, the TACCESS papers' title is TACCESSVolumelIssue-Number of Article, e.g. TACCESS1201-03. Keywords are also missing in the TACCESS publications' metadata. Additionally, numerous PDF files were not text-searchable. For ASSETS, these were five publications. While this resembles only 1.6%, this could still have biased our results.

6 CONCLUSION

In conclusion, we have shown that people with disabilities are addressed in numerous conferences in HCI. Some of the venues seem to address specific disabilities disproportionately. ASSETS and TACCESS, for example, seem to mainly address visual impairments. To uncover this finding, we employed a script-based methodology. We proposed some explanations such that *visual disabilities* are more graspable for researchers and practitioners. Our work is, to the best of our knowledge, the first approach to quantitatively reveal the addressed populations with disabilities in HCI.

ACKNOWLEDGMENTS

This work was conducted within the project 'SituWare' funded by the Federal Ministry for Economic Affairs and Energy (BMWi). Mark Colley was supported by the Startup Funding B of Ulm University.

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A SEARCH TERMS

Vision Disability: 'Braille-Reading', 'Impaired Colour Vision', 'Tactile Media', 'functional vision', 'blindness', 'Vision Disability', 'visually impaired', 'visual impairment', 'Visual Impairments', 'visually', 'impaired vision', 'braille', 'visually handicapped', 'low vision', 'low-vision', 'blind', 'visually-impaired', 'blind-folded', 'People with Vision Impairments', 'Accessible Image', 'accessible EPUB3', 'non-visual interaction', 'partially sighted', 'visual disability', 'visually challenged', 'poor eyesight', 'near-blind', 'dim-sighted', 'visual-to-auditory', 'Visual Disabilities', 'cane', 'image text alternatives', 'Aural Navigation', 'image text alternative', 'visual handicap', 'short-sighted', 'Technology-Mediated Sight', 'VIP', 'tunnel vision simulation', 'Screen-Free', 'Screenless', 'Auditory Interaction', 'Synesthesia vision', 'Audio-Tactile Web Browsing', 'colour vision deficiency', 'Auditory Graphing Software', 'Color Differentiation', 'Color vision', 'screen-reader', 'Screen Reader', 'tactile graphics', 'haptic graphics', 'Mixed Visual Abilities', 'Visual Accessibility', 'Foresee', 'Tesseract OCR', 'Accessible Map Visualization', 'Tangram', 'Refractive error', 'ocular aberration'

Hearing Disability: 'Hard-of-Hearing', 'American Sign Language', 'Sign Language', 'subtitle', 'Mobile Sign Language', 'Hearing Disability', 'deaf', 'hearing loss', 'impaired hearing', 'hearing impaired', 'Hearing Disabilities', 'hearing-impaired', 'deafness', 'hard of hearing', 'impaired hearing', 'hearing disorder', 'hypoacusis', 'hearing difficulties', 'ASL', 'real-time captioning', 'hearing problem', 'loss of hearing', 'Lip-reading', 'Hearing Aid', 'Cochlear Implants', 'Speechreading', 'divergent hearing', 'transcription'

Brain, Cognitive or Learning Disability: 'Inclusive Education', 'Inclusion and Education', 'Speech-Language Therapy', 'intellectual disabilities', 'cognitive impairment', 'cognitively impaired', 'Developmental Disabilities', 'Developmental Disability', 'Brain Injury', 'Mental Ill-health', 'learning disabilities', 'learning disability', 'Developmental Disabilities', 'Mental Disabilities', 'Mental Disability', 'Cognitive Disabilities', 'Cognitive Disability', 'Cognitively Impaired', 'Cognitive or Learning Disabilities', 'dyslexia', 'cognitive disorder', 'cognitive impairment', 'learning difficulty', 'learning difficulties', 'learning-disabled', 'intellectual disability', 'intellectual', 'Cognitive Assistance', 'text simplification', 'dyslexic', 'learning disorder', 'down syndrome', 'speech impairment', 'non-speaking', 'non-speaking', 'speech-language therapy', 'Complex Communication Needs', 'Limited Communication', 'SimpleNLG', 'Italian Dysarthric Speech', 'Dysarthria', 'stroke', 'Apoplexy', 'Brain Disability', 'Brain Disabilities', 'brain', 'Acquired Brain Injury', 'Traumatic Brain Injury', 'brain disease', 'encephalopathy', 'brain problem', 'brain condition', 'brain disorder', 'alzheimer', 'Parkinson', 'with PD', 'Amnesia', 'dementia', 'verbal dyspraxia', 'apraxia', 'childhood apraxia of speech'

Psychological Disability: 'Psychiatric test', 'Borderline', 'Dialectical Behavioral', 'Psychological Disorders', 'schizophrenia', 'individuals with SZ', 'neurotypical', 'personality', 'psychological problem', 'psychological distress', 'psychological disturbance', 'mental disorder', 'mental health', 'mental problems', 'mental illness', 'mental disturbance', 'folie', 'psychic disorder', 'neurosis', 'psychological condition', 'eating disorder', 'anxiety disorder', 'Invisible Disability', 'Invisible Disabilities', 'adhd', 'epilepsy', 'chronic pain', 'chronic fatigue', 'chronic dizziness', 'anxiety disorder', 'allergy', 'arachnoiditis', 'asperger', 'asthma', 'autism', 'with ASD', 'bipolar disorder', 'charcot-marie-tooth', 'circadian rhythm', 'coeliac', 'crohn', 'depression', 'diabetes', 'ehlers', 'endometriosis', 'fetal alcohol', 'Fibromyalgia', 'magraine', 'multiple sclerosis', 'narcolepsy', 'repetitive stress', 'scleroderma', 'aphasia', 'AAC', 'Autistic Spectrum Disorder', 'Childhood psychosis'

Mobility Impairment: 'Spinal Cord Disability', 'Poliomyelitis', 'Sidewalk Accessibility', 'gait', 'Mobility Analysis', 'motor impairment', 'motor disability', 'motor disabilities', 'wheelchair', 'wheelchair-based', 'hand tremor', 'tremor', 'Cerebral palsy', 'Muscular dystrophy', 'carpal tunnel', 'arthritis', 'Dexterity', 'Mobility and Physical Impairments', 'mobility impairment', 'physical handicap', 'physically disabled', 'physically handicapped', 'physical inability', 'Neuromuscular', 'hemiparesis', 'Variable Pointing Performance', 'gaze-control', 'Upper Limb Rehabilitation', 'Friedreich', 'ataxia', 'wearable', 'physical condition', 'tetraplegic', 'motor-impaired', 'motor impaired', 'limited motor abilities', 'physical impairment', 'physically disabled'

Elderly: 'Osteoporosis', 'older adult', 'Ageing', 'Aging', 'Elderly',

'care-giving environment', 'Elder', 'Senior-Friendly', 'senior persona', 'Elder Connectedness', 'older', 'mature', 'senior', 'elder', 'older person', 'alzheimer', 'seniors', 'social isolation', 'loneliness', 'retirement residence', 'Independent Living'

General: 'universal access', 'accessibility barriers', 'universally accessible', 'With Disabilities', 'with disability', 'Accessible Local Government', 'Web Design Guidelines', 'Teaching Accessibility', 'Accessible learning', 'Accessible Video Player', 'Accessible OzPlayer', 'Accessibility Guidelines', 'Disabled', 'Accessible Statistics', 'Inclusive Web', 'accessibility education', 'accessibility policies', 'website accessibility', 'web accessibility', 'website accessibility', 'Social Connectedness', 'digital accessibility', 'social accessibility', 'App Accessibility', 'Chronic Conditions', 'Diverse Needs', 'Cancer', 'Printing Assistive Technology', 'in Accessibility Research', 'Assistive Technology', 'Disability and Technology', 'General Accessibility', 'Impairment Simulation', 'Environmental Accessibility', 'Special Education', 'special support needs', 'special needs', 'Assistive Services', 'Situational impairment', 'accessible PDF', 'STEM Accessibility', 'Web Content Modality', 'Capti ESL Assistant', 'disability studies', 'Assisted Living', 'Living', 'for the disabled', 'Accessible Interactive Simulations', 'Accessibility Evaluation', 'Accessibility Policies', 'Situational Impairment', 'Accessibility Feature'

Table 3: Distances between manual and automatic categorization for ASSETS.

Year	Vision	Hearing	Brain, Cognitive, or Learning	Psychological	Mobility and Psychological	General	Elderly
2009	2	0	0	0	0	0	0
2010	1	0	0	0	1	0	0
2011	1	0	0	1	1	1	0
2012	2	0	0	0	0	0	0
2013	0	0	0	0	0	0	1
2014	0	0	0	0	0	0	0
2015	2	0	1	0	1	0	1
2016	0	0	0	0	0	0	0
2017	1	0	0	0	0	1	0
2018	0	0	1	0	2	1	0
2019	0	0	0	0	0	0	0
Combined	9	0	2	1	5	3	2

B METHOD

C R SCRIPT

```

1 library(pacman)
2
3 pacman::p_load(pastecs, readxl, xlsx, pdfsearch,
4               tabulizer, formattable, htmltools, webshot, DT,
5               Rfast, ggplot2, beeper)
6
7 # outsource all definitions for better overview
8 source("PATH_TO_DEFINITIONS.R", echo = FALSE)
9
10 number_of_problems <- 0
11 number_of_not_categorized <- 0
12
13 potentially_related_work <- 0
14 unrelated_work <- 0
15
16 categorized_by_title <- 0
17 categorized_by_keywords <- 0
18
19 df_occurrences <- data.frame(
20   Conference = character(),
21   Vision = integer(),
22   Hearing = integer(),
23   Cognitive = integer(),
24   Psychological = integer(),
25   Mobility = integer(),
26   General = integer(),
27   Elderly = integer(),
28   stringsAsFactors = FALSE
29 )
30
31 df_occurrences_un_categorized <- data.frame(
32   Conference = character(),
33   Vision = integer(),
34   Hearing = integer(),
35   Cognitive = integer(),
36   Psychological = integer(),
37   Mobility = integer(),
38   General = integer(),
39   Elderly = integer(),
40   stringsAsFactors = FALSE
41 )

```

```

42 df_accessibility_related_keywords_found <- data.frame(
43   Conference = character(),
44   number = integer(),
45   stringsAsFactors = FALSE
46 )
47
48 # Set directory to get all relevant pdfs, do this for all
49   years
50 setwd("PATH_FOR_FILES_TOP_FOLDER")
51
52 # get all conference directories
53 conference_directories <- list.dirs(".", recursive =
54   FALSE)
55
56 sink("output_pdf_search_newest.txt")
57 sink(stdout(), type = "message")
58
59 for (d in 1:length(conference_directories)) {
60
61   # go into x'th conference
62   setwd(conference_directories[d])
63
64   df_conference <- data.frame(
65     Year = character(),
66     "Vision Disability" = integer(),
67     "Hearing Disability" = integer(),
68     "Cognitive Disability" = integer(),
69     "Psychological Disability" = integer(),
70     "Mobility Disability" = integer(),
71     "General Disability" = integer(),
72     "Elderly" = integer(),
73     stringsAsFactors = FALSE
74   )
75
76 # get all years
77 year_directories <- list.dirs(".", recursive = FALSE)
78
79
80
81 for (y in 1:length(year_directories)) {
82   vision_related_publications <- 0
83   hearing_related_publications <- 0
84   cognitive_related_publications <- 0
85   psychological_related_publications <- 0
86   mobility_related_publications <- 0
87   general_related_publications <- 0
88   elderly_related_publications <- 0
89
90   # go into x'th year

```

```

92   setwd(year_directories[y])
93
94   year <- as.integer(substring(year_directories[y], 3))
95
96   files <- list.files(pattern = "\\pdf$")
97
98   for (i in 1:length(files)) {
99
100     # message all the names of the files
101     # message(i, ":", substr(files[i], 1, nchar(
102       files[i])-4))
103
104     # pdf_file <- system.file('pdf', files[i], package
105       = 'pdfsearch')
106     # The pdftools function for extracting text is pdf_
107     text.
108     pdf_file <- files[i]
109
110     result <- read_Metadate(pdf_file = pdf_file)
111     title_available <- TRUE
112
113     # Title was extracted
114     # Attention: result if gone right is an array of
115     # relevant information including author, title
116     # if gone wrong, then it is simply ""
117     # Metadata can be available but title can still be
118     NULL!
119     # Use '||' to avoid error: "$ operator is invalid
120     # for atomic vectors"
121     # see https://stat.ethz.ch/R-manual/R-devel/library
122     /base/html/Logic.html
123     if (length(result) == 1 || is.null(result$title)) {
124       message("Title of PDF NOT AVAILABLE")
125       title_available <- FALSE
126     } else {
127       message("Title of PDF is:", result$title)
128     }
129
130     # we search in the title and if any keyword was
131     found, we assume that this is the disability studied
132     # we once don't look at this to get information on
133     the distribution!
134     if (title_available) {
135       # message("Looking in TITLE")
136
137       pdf_title <- result$title

```

```

134 result_disability_vision <- keyword_search(pdf_
135 title,
136 keyword = keyword_disability_vision,
137 path = FALSE,
138 ignore_case = TRUE, split_pdf = FALSE
139 )
140 result_disability_hearing <- keyword_search(pdf_
141 title,
142 keyword = keyword_disability_hearing,
143 path = FALSE,
144 ignore_case = TRUE, split_pdf = FALSE
145 )
146 result_disability_cognitive <- keyword_search(pdf_
147 title,
148 keyword = keyword_disability_cognitive,
149 path = FALSE,
150 ignore_case = TRUE, split_pdf = FALSE
151 )
152 result_disability_psychological <- keyword_search
153 (pdf_title,
154 keyword = keyword_disability_psychological,
155 path = FALSE,
156 ignore_case = TRUE, split_pdf = FALSE
157 )
158 result_disability_mobility <- keyword_search(pdf_
159 title,
160 keyword = keyword_disability_mobility,
161 path = FALSE,
162 ignore_case = TRUE, split_pdf = FALSE
163 )
164 result_disability_older <- keyword_search(pdf_
165 title,
166 keyword = keyword_disability_older,
167 path = FALSE,
168 ignore_case = TRUE, split_pdf = FALSE
169 )
170 result_disability_general <- keyword_search(pdf_
171 title,
172 keyword = keyword_disability_general,
173 path = FALSE,
174 ignore_case = TRUE, split_pdf = FALSE
175 )
176 # anyone with a keyword?
177
178 if (dim(result_disability_vision)[1] > 0) {
179   # a term was found!
180   message("Vision disability term was found in
181 TITLE. Number:", dim(result_disability_vision)[1])
182   # message(head(result_disability_vision))
183   vision_related_publications <- vision_related_
184 publications + 1
185   categorized_by_title <- categorized_by_title +
186 1
187   # next to avoid double counting!
188   # but only next when there really is one,
189   otherwise we look in entire document
190   next
191 }
192
193 if (dim(result_disability_hearing)[1] > 0) {
194   # a term was found in TITLE!
195   message("Hearing disability term was found in
196 TITLE. Number:", dim(result_disability_hearing)[1])
197   # message(head(result_disability_hearing))
198   hearing_related_publications <- hearing_related_
199 publications + 1
200   categorized_by_title <- categorized_by_title +
201 1
202   next
203 }

```

```

198 if (dim(result_disability_cognitive)[1] > 0) {
199   # a term was found in TITLE!
200   message("Cognitive disability term was found in
201 TITLE. Number:", dim(result_disability_cognitive)
202 [1])
203   # message(head(result_disability_cognitive))
204   cognitive_related_publications <- cognitive_
205 related_publications + 1
206   categorized_by_title <- categorized_by_title +
207 1
208   next
209 }
210
211 if (dim(result_disability_psychological)[1] > 0)
212 {
213   # a term was found in TITLE!
214   message("Psychological disability term was
215 found in TITLE. Number:", dim(result_disability_
216 psychological)[1])
217   # message(head(result_disability_psychological)
218 )
219   psychological_related_publications <-
220 psychological_related_publications + 1
221   categorized_by_title <- categorized_by_title +
222 1
223   next
224 }
225
226 if (dim(result_disability_mobility)[1] > 0) {
227   # a term was found in TITLE!
228   message("Mobility disability term was found in
229 TITLE. Number:", dim(result_disability_mobility)[1])
230   # message(head(result_disability_mobility))
231   mobility_related_publications <- mobility_
232 related_publications + 1
233   categorized_by_title <- categorized_by_title +
234 1
235   next
236 }
237
238 if (dim(result_disability_older)[1] > 0) {
239   # a term was found in TITLE!
240   message("Elderly disability term was found in
241 TITLE. Number:", dim(result_disability_older)[1])
242   # message(head(result_disability_older))
243   elderly_related_publications <- elderly_related_
244 publications + 1
245   categorized_by_title <- categorized_by_title +
246 1
247   next
248 }
249
250 if (dim(result_disability_general)[1] > 0) {
251   # a term was found in TITLE!
252   message("General disability term was found in
253 TITLE. Number:", dim(result_disability_general)[1])
254   # message(head(result_disability_general))
255   general_related_publications <- general_related_
256 publications + 1
257   categorized_by_title <- categorized_by_title +
258 1
259   next
260 }
261 }
262
263 keywords_available <- TRUE
264
265 # keywords_available was extracted
266 # Attention: result if gone right is an array of
267 relevant information including author, title
268 # if gone wrong, then it is simply ""
269 # see https://stat.ethz.ch/R-manual/R-devel/library
270 /base/html/Logic.html

```

```

254   if (length(result) == 1 || is.null(result$keywords)
    ) {
256     message("Keywords of PDF NOT AVAILABLE")
      keywords_available <- FALSE
    } else {
258     message("Keywords are:", result$keywords)
    }
260
262   # we search in the keywords and if any keyword was
    found, we assume that this is the disability studied
    # we once don't look at this to get information on
    the distribution!
264   if (keywords_available) {
      # message("Looking in KEYWORDS")
266
      pdf_keywords <- result$keywords
268
      result_disability_vision <- keyword_search(pdf_
keywords,
270       keyword = keyword_disability_vision,
        path = FALSE,
272       ignore_case = TRUE, split_pdf = FALSE
    )
274
      result_disability_hearing <- keyword_search(pdf_
keywords,
276       keyword = keyword_disability_hearing,
        path = FALSE,
278       ignore_case = TRUE, split_pdf = FALSE
    )
280
      result_disability_cognitive <- keyword_search(pdf_
keywords,
282       keyword = keyword_disability_cognitive,
        path = FALSE,
284       ignore_case = TRUE, split_pdf = FALSE
    )
286
      result_disability_psychological <- keyword_search
(pdf_keywords,
288       keyword = keyword_disability_psychological,
        path = FALSE,
290       ignore_case = TRUE, split_pdf = FALSE
    )
292
      result_disability_mobility <- keyword_search(pdf_
keywords,
294       keyword = keyword_disability_mobility,
        path = FALSE,
296       ignore_case = TRUE, split_pdf = FALSE
    )
298
      result_disability_older <- keyword_search(pdf_
keywords,
300       keyword = keyword_disability_older,
        path = FALSE,
302       ignore_case = TRUE, split_pdf = FALSE
    )
304
      result_disability_general <- keyword_search(pdf_
keywords,
306       keyword = keyword_disability_general,
        path = FALSE,
308       ignore_case = TRUE, split_pdf = FALSE
    )
310
312   # anyone with a keyword?
314
316   if (dim(result_disability_vision)[1] > 0) {
      # a term was found!
      message("Vision disability term was found in
KEYWORDS Number:", dim(result_disability_vision)[1])
      # message(head(result_disability_vision))
318     vision_related_publications <- vision_related_
publications + 1

```

```

      categorized_by_keywords <- categorized_by_
keywords + 1
      # next to avoid double counting!
      # but only next when there really is one,
      otherwise we look in entire document
      next
    }
324
326   if (dim(result_disability_hearing)[1] > 0) {
      message("Hearing disability term was found in
KEYWORDS Number:", dim(result_disability_hearing)
[1])
      # message(head(result_disability_hearing))
      hearing_related_publications <- hearing_related_
publications + 1
      categorized_by_keywords <- categorized_by_
keywords + 1
      next
    }
332
334   if (dim(result_disability_cognitive)[1] > 0) {
      message("Cognitive disability term was found in
KEYWORDS Number:", dim(result_disability_cognitive)
[1])
      # message(head(result_disability_cognitive))
      cognitive_related_publications <- cognitive_
related_publications + 1
      categorized_by_keywords <- categorized_by_
keywords + 1
      next
    }
340
342   if (dim(result_disability_psychological)[1] > 0)
    {
      message("Psychological disability term was
found in KEYWORDS Number:", dim(result_disability_
psychological)[1])
      # message(head(result_disability_psychological)
)
      psychological_related_publications <-
psychological_related_publications + 1
      categorized_by_keywords <- categorized_by_
keywords + 1
      next
    }
348
350   if (dim(result_disability_mobility)[1] > 0) {
      message("Mobility disability term was found in
KEYWORDS Number:", dim(result_disability_mobility)
[1])
      # message(head(result_disability_mobility))
      mobility_related_publications <- mobility_
related_publications + 1
      categorized_by_keywords <- categorized_by_
keywords + 1
      next
    }
356
358   if (dim(result_disability_older)[1] > 0) {
      message("Elderly disability term was found in
KEYWORDS Number:", dim(result_disability_older)[1])
      # message(head(result_disability_older))
      elderly_related_publications <- elderly_related_
publications + 1
      categorized_by_keywords <- categorized_by_
keywords + 1
      next
    }
364
366   if (dim(result_disability_general)[1] > 0) {
      message("General disability term was found in
KEYWORDS Number:", dim(result_disability_general)
[1])
      # message(head(result_disability_general))
      general_related_publications <- general_related_
publications + 1

```

```

370     categorized_by_keywords <- categorized_by_
keywords + 1
372     next
374   }
376   # check for each keyword for the disability
377   if (TRUE) {
378     result_disability_vision <- keyword_search(pdf_
file ,
380     keyword = keyword_disability_vision ,
path = TRUE, remove_hyphen = TRUE, surround_
lines = 1,
382     ignore_case = TRUE, split_pdf = TRUE
)
384     result_disability_hearing <- keyword_search(pdf_
file ,
386     keyword = keyword_disability_hearing ,
path = TRUE, remove_hyphen = TRUE, surround_
lines = 1,
388     ignore_case = TRUE, split_pdf = TRUE
)
390     result_disability_cognitive <- keyword_search(pdf_
file ,
392     keyword = keyword_disability_cognitive ,
path = TRUE, remove_hyphen = TRUE, surround_
lines = 1,
394     ignore_case = TRUE, split_pdf = TRUE
)
396     result_disability_psychological <- keyword_search
(pdf_file ,
398     keyword = keyword_disability_psychological ,
path = TRUE, remove_hyphen = TRUE, surround_
lines = 1,
400     ignore_case = TRUE, split_pdf = TRUE
)
402     result_disability_mobility <- keyword_search(pdf_
file ,
404     keyword = keyword_disability_mobility ,
path = TRUE, remove_hyphen = TRUE, surround_
lines = 1,
406     ignore_case = TRUE, split_pdf = TRUE
)
408     result_disability_older <- keyword_search(pdf_
file ,
410     keyword = keyword_disability_older ,
path = TRUE, remove_hyphen = TRUE, surround_
lines = 1,
412     ignore_case = TRUE, split_pdf = TRUE
)
414     result_disability_general <- keyword_search(pdf_
file ,
416     keyword = keyword_disability_general ,
path = TRUE, remove_hyphen = TRUE, surround_
lines = 1,
418     ignore_case = TRUE, split_pdf = TRUE
)
420   }
422   # CHECK for which criteria is met -- TRUE to always
do it
424   if (TRUE) {
426     # create row for all found keywords
df_one_year <- data.frame(
428     Conference = substring(conference_directories[d
], 3),
Vision = dim(result_disability_vision)[1],
Hearing = dim(result_disability_hearing)[1],

```

```

430     Cognitive = dim(result_disability_cognitive)
[1],
432     Psychological = dim(result_disability_
psychological)[1],
Mobility = dim(result_disability_mobility)[1],
434     General = dim(result_disability_general)[1],
Elderly = dim(result_disability_older)[1],
436     stringsAsFactors = FALSE
)
438     # string these together
df_occurrences <- rbind(df_occurrences , df_one_
year)
440
442     # now get total number of found acc. related
keywords
accessibility_related_occurrences <- dim(result_
disability_vision)[1] + dim(result_disability_
hearing)[1] + dim(result_disability_cognitive)[1] +
444     dim(result_disability_general)[1] + dim(result_
disability_hearing)[1] + dim(result_disability_
mobility)[1] + dim(result_disability_older)[1] + dim
(result_disability_psychological)[1]
446     message("Number of accessibility related
occurrences: ", accessibility_related_occurrences)
448
450     # look how many accessibility_related_occurrences
are found in paper
df_accessibility_found_one_year <- data.frame(
452     Conference = substring(conference_directories[d
], 3),
number = accessibility_related_occurrences ,
454     stringsAsFactors = FALSE
)
456
458     # string these together
df_accessibility_related_keywords_found <- rbind(
460     df_accessibility_related_keywords_found, df_
accessibility_found_one_year)
462
464     if (dim(result_disability_vision)[1] > MINIMUM_
OCCURRENCES_FOR_ACCEPTANCE) {
466       # compute all ratios
# ratioVisionHearing <- dim(result_disability_
vision)[1] / dim(result_disability_hearing)[1]
# ratioVisionCognitive <- dim(result_disability_
vision)[1] / dim(result_disability_cognitive)[1]
468     # ratioVisionPsychological <- dim(result_
disability_vision)[1] / dim(result_disability_
psychological)[1]
# ratioVisionMobility <- dim(result_disability_
vision)[1] / dim(result_disability_mobility)[1]
# ratioVisionOlder <- dim(result_disability_
vision)[1] / dim(result_disability_older)[1]
# ratioVisionGeneral <- dim(result_disability_
vision)[1] / dim(result_disability_general)[1]
470     secondHighest <- order(dim(result_disability_
hearing)[1], dim(result_disability_cognitive)[1],
dim(result_disability_psychological)[1], dim(result_
disability_mobility)[1], dim(result_disability_older
)[1], dim(result_disability_general)[1])
ratioVisionsecondHighest <- dim(result_
disability_vision)[1] / secondHighest
472
474     # if(ratioVisionHearing > MINIMUM_RATIO &
ratioVisionCognitive > MINIMUM_RATIO &
ratioVisionPsychological > MINIMUM_RATIO &
ratioVisionMobility > MINIMUM_RATIO &
ratioVisionOlder > MINIMUM_RATIO &
ratioVisionGeneral > MINIMUM_RATIO){

```

```

    if (ratioVisionsecondHighest > MINIMUM_RATIO |
ratioVisionsecondHighest == Inf) {
    vision_related_publications <- vision_related
_publications + 1
    # here we go to next one as this is already
categorized
    next
}

# now check that all others are below threshold
else if ((dim(result_disability_hearing)[1] >
MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
disability_cognitive)[1] > MAXIMUM_OCCURRENCES_FOR_
MANUAL) | (dim(result_disability_psychological)[1] >
MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
disability_mobility)[1] > MAXIMUM_OCCURRENCES_FOR_
MANUAL) | (dim(result_disability_older)[1] > MAXIMUM_
_OCCURRENCES_FOR_MANUAL) | (dim(result_disability_
general)[1] > MAXIMUM_OCCURRENCES_FOR_MANUAL)) {
    message("Error for
paper, manual adjustment needed: ")
    message(" ", files[i])
    # message(" ", metadata$
title))
    number_of_problems <- number_of_problems + 1
} else {
    # otherwise add one to the year and the
correct column
    vision_related_publications <- vision_related
_publications + 1
    # here we go to next one as this is already
categorized
    next
} else if (dim(result_disability_hearing)[1] >
MINIMUM_OCCURRENCES_FOR_ACCEPTANCE) {

    # compute all ratios
    # ratioHearingVision <- dim(result_disability_
hearing)[1] / dim(result_disability_vision)[1]
    # ratioHearingCognitive <- dim(result_
disability_hearing)[1] / dim(result_disability_
cognitive)[1]
    # ratioHearingPsychological <- dim(result_
disability_hearing)[1] / dim(result_disability_
psychological)[1]
    # ratioHearingMobility <- dim(result_disability_
_hearing)[1] / dim(result_disability_mobility)[1]
    # ratioHearingOlder <- dim(result_disability_
hearing)[1] / dim(result_disability_older)[1]
    # ratioHearingGeneral <- dim(result_disability_
hearing)[1] / dim(result_disability_general)[1]

    secondHighest <- order(dim(result_disability_
vision)[1], dim(result_disability_cognitive)[1], dim
(result_disability_psychological)[1], dim(result_
disability_mobility)[1], dim(result_disability_older
)[1], dim(result_disability_general)[1])
    ratioHearingsecondHighest <- dim(result_
disability_hearing)[1] / secondHighest

    # if(ratioHearingVision > MINIMUM_RATIO &
ratioHearingCognitive > MINIMUM_RATIO &
ratioHearingPsychological > MINIMUM_RATIO &
ratioHearingMobility > MINIMUM_RATIO &
ratioHearingOlder > MINIMUM_RATIO &
ratioHearingGeneral > MINIMUM_RATIO){
    if (ratioHearingsecondHighest > MINIMUM_RATIO |
ratioHearingsecondHighest == Inf) {
    hearing_related_publications <- hearing_
related_publications + 1
    # here we go to next one as this is already
categorized
    next
}
}

```

```

    # now check that all others are below threshold
else if ((dim(result_disability_vision)[1] >
MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
disability_cognitive)[1] > MAXIMUM_OCCURRENCES_FOR_
MANUAL) | (dim(result_disability_psychological)[1] >
MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
disability_mobility)[1] > MAXIMUM_OCCURRENCES_FOR_
MANUAL) | (dim(result_disability_older)[1] > MAXIMUM_
_OCCURRENCES_FOR_MANUAL) | (dim(result_disability_
general)[1] > MAXIMUM_OCCURRENCES_FOR_MANUAL)) {
    message("Error for
paper, manual adjustment needed: ")
    message(" ", files[i])
    # message(" ", metadata$
title))
    number_of_problems <- number_of_problems + 1
} else {
    # otherwise add one to the year and the
correct column
    hearing_related_publications <- hearing_
related_publications + 1
    next
} else if (dim(result_disability_cognitive)[1] >
MINIMUM_OCCURRENCES_FOR_ACCEPTANCE) {

    # compute all ratios
    # ratioCognitiveHearing <- dim(result_
disability_cognitive)[1] / dim(result_disability_
hearing)[1]
    # ratioCognitiveVision <- dim(result_disability_
_cognitive)[1] / dim(result_disability_vision)[1]
    # ratioCognitivePsychological <- dim(result_
disability_cognitive)[1] / dim(result_disability_
psychological)[1]
    # ratioCognitiveMobility <- dim(result_
disability_cognitive)[1] / dim(result_disability_
mobility)[1]
    # ratioCognitiveOlder <- dim(result_disability_
cognitive)[1] / dim(result_disability_older)[1]
    # ratioCognitiveGeneral <- dim(result_
disability_cognitive)[1] / dim(result_disability_
general)[1]

    secondHighest <- order(dim(result_disability_
hearing)[1], dim(result_disability_vision)[1], dim(
result_disability_psychological)[1], dim(result_
disability_mobility)[1], dim(result_disability_older
)[1], dim(result_disability_general)[1])
    ratioCognitivesecondHighest <- dim(result_
disability_cognitive)[1] / secondHighest

    # if(ratioCognitiveHearing > MINIMUM_RATIO &
ratioCognitiveVision > MINIMUM_RATIO &
ratioCognitivePsychological > MINIMUM_RATIO &
ratioCognitiveMobility > MINIMUM_RATIO &
ratioCognitiveOlder > MINIMUM_RATIO &
ratioCognitiveGeneral > MINIMUM_RATIO){
    if (ratioCognitivesecondHighest > MINIMUM_RATIO
| ratioCognitivesecondHighest == Inf) {
    cognitive_related_publications <- cognitive_
related_publications + 1
    # here we go to next one as this is already
categorized
    next
}
}

    # now check that all others are below threshold

```

```

else if ((dim(result_disability_vision)[1] >
MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
disability_hearing)[1] > MAXIMUM_OCCURRENCES_FOR_
MANUAL) | (dim(result_disability_psychological)[1] >
MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
disability_mobility)[1] > MAXIMUM_OCCURRENCES_FOR_
MANUAL) | (dim(result_disability_older)[1] > MAXIMUM_
_OCCURRENCES_FOR_MANUAL) | (dim(result_disability_
general)[1] > MAXIMUM_OCCURRENCES_FOR_MANUAL)) {
message("Error for
paper, manual adjustment needed: ")
message(" ", files[i])
# message(" ", metadata$
title))
number_of_problems <- number_of_problems + 1
} else {
# otherwise add one to the year and the
correct column
cognitive_related_publications <- cognitive_
related_publications + 1
next
} else if (dim(result_disability_psychological)
[1] > MINIMUM_OCCURRENCES_FOR_ACCEPTANCE) {

# compute all ratios
# ratioPsychologicalHearing <- dim(result_
disability_psychological)[1] / dim(result_disability_
hearing)[1]
# ratioPsychologicalCognitive <- dim(result_
disability_psychological)[1] / dim(result_disability_
_cognitive)[1]
# ratioPsychologicalVision <- dim(result_
disability_psychological)[1] / dim(result_disability_
_vision)[1]
# ratioPsychologicalMobility <- dim(result_
disability_psychological)[1] / dim(result_disability_
_mobility)[1]
# ratioPsychologicalOlder <- dim(result_
disability_psychological)[1] / dim(result_disability_
_older)[1]
# ratioPsychologicalGeneral <- dim(result_
disability_psychological)[1] / dim(result_disability_
_general)[1]

secondHighest <- order(dim(result_disability_
hearing)[1], dim(result_disability_cognitive)[1],
dim(result_disability_vision)[1], dim(result_
disability_mobility)[1], dim(result_disability_older
)[1], dim(result_disability_general)[1])
ratioPsychologicalsecondHighest <- dim(result_
disability_psychological)[1] / secondHighest

# now check that all others are below threshold
# if(ratioPsychologicalHearing > MINIMUM_RATIO
& ratioPsychologicalCognitive > MINIMUM_RATIO &
ratioPsychologicalVision > MINIMUM_RATIO &
ratioPsychologicalMobility > MINIMUM_RATIO &
ratioPsychologicalOlder > MINIMUM_RATIO &
ratioPsychologicalGeneral > MINIMUM_RATIO){
if (ratioPsychologicalsecondHighest > MINIMUM_
RATIO | ratioPsychologicalsecondHighest == Inf) {
psychological_related_publications <-
psychological_related_publications + 1
# here we go to next one as this is already
categorized
next
}
}

# now check that all others are below threshold

```

```

else if ((dim(result_disability_vision)[1] >
MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
disability_cognitive)[1] > MAXIMUM_OCCURRENCES_FOR_
MANUAL) | (dim(result_disability_hearing)[1] >
MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
disability_mobility)[1] > MAXIMUM_OCCURRENCES_FOR_
MANUAL) | (dim(result_disability_older)[1] > MAXIMUM_
_OCCURRENCES_FOR_MANUAL) | (dim(result_disability_
general)[1] > MAXIMUM_OCCURRENCES_FOR_MANUAL)) {
message("Error for
paper, manual adjustment needed: ")
message(" ", files[i])
# message(" ", metadata$
title))
number_of_problems <- number_of_problems + 1
} else {
# otherwise add one to the year and the
correct column
psychological_related_publications <-
psychological_related_publications + 1
next
} else if (dim(result_disability_mobility)[1] >
MINIMUM_OCCURRENCES_FOR_ACCEPTANCE) {

# compute all ratios
# ratioMobilityHearing <- dim(result_disability_
_mobility)[1] / dim(result_disability_hearing)[1]
# ratioMobilityCognitive <- dim(result_
disability_mobility)[1] / dim(result_disability_
_cognitive)[1]
# ratioMobilityPsychological <- dim(result_
disability_mobility)[1] / dim(result_disability_
_psychological)[1]
# ratioMobilityVision <- dim(result_disability_
_mobility)[1] / dim(result_disability_vision)[1]
# ratioMobilityOlder <- dim(result_disability_
_mobility)[1] / dim(result_disability_older)[1]
# ratioMobilityGeneral <- dim(result_disability_
_mobility)[1] / dim(result_disability_general)[1]

secondHighest <- order(dim(result_disability_
hearing)[1], dim(result_disability_cognitive)[1],
dim(result_disability_psychological)[1], dim(result_
disability_vision)[1], dim(result_disability_older)
[1], dim(result_disability_general)[1])
ratioMobilitysecondHighest <- dim(result_
disability_mobility)[1] / secondHighest

# if(ratioMobilityHearing > MINIMUM_RATIO &
ratioMobilityCognitive > MINIMUM_RATIO &
ratioMobilityPsychological > MINIMUM_RATIO &
ratioMobilityVision > MINIMUM_RATIO &
ratioMobilityOlder > MINIMUM_RATIO &
ratioMobilityGeneral > MINIMUM_RATIO){
if (ratioMobilitysecondHighest > MINIMUM_RATIO
| ratioMobilitysecondHighest == Inf) {
mobility_related_publications <- mobility_
related_publications + 1
# here we go to next one as this is already
categorized
next
}
}

# now check that all others are below threshold
else if ((dim(result_disability_vision)[1] >
MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
disability_cognitive)[1] > MAXIMUM_OCCURRENCES_FOR_
MANUAL) | (dim(result_disability_psychological)[1] >
MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
disability_hearing)[1] > MAXIMUM_OCCURRENCES_FOR_
MANUAL) | (dim(result_disability_older)[1] > MAXIMUM_
_OCCURRENCES_FOR_MANUAL) | (dim(result_disability_
general)[1] > MAXIMUM_OCCURRENCES_FOR_MANUAL)) {
message("Error for
paper, manual adjustment needed: ")

```

```

        message(" ", files[i])
        # message(" ", metadata$title))
        number_of_problems <- number_of_problems + 1
      } else {
        # otherwise add one to the year and the
        correct column
        mobility_related_publications <- mobility_
        related_publications + 1
        next
      } else if (dim(result_disability_older)[1] >
        MINIMUM_OCCURRENCES_FOR_ACCEPTANCE) {

        # compute all ratios
        # ratioOlderHearing <- dim(result_disability_
        older)[1] / dim(result_disability_hearing)[1]
        # ratioOlderCognitive <- dim(result_disability_
        older)[1] / dim(result_disability_cognitive)[1]
        # ratioOlderPsychological <- dim(result_
        disability_older)[1] / dim(result_disability_
        psychological)[1]
        # ratioOlderMobility <- dim(result_disability_
        older)[1] / dim(result_disability_mobility)[1]
        # ratioOlderVision <- dim(result_disability_
        older)[1] / dim(result_disability_vision)[1]
        # ratioOlderGeneral <- dim(result_disability_
        older)[1] / dim(result_disability_general)[1]

        secondHighest <- order(dim(result_disability_
        hearing)[1], dim(result_disability_cognitive)[1],
        dim(result_disability_psychological)[1], dim(result_
        disability_mobility)[1], dim(result_disability_
        vision)[1], dim(result_disability_general)[1])
        ratioOldersecondHighest <- dim(result_disability_
        older)[1] / secondHighest

        # if(ratioOlderHearing > MINIMUM_RATIO &
        ratioOlderCognitive > MINIMUM_RATIO &
        ratioOlderPsychological > MINIMUM_RATIO &
        ratioOlderMobility > MINIMUM_RATIO &
        ratioOlderVision > MINIMUM_RATIO & ratioOlderGeneral
        > MINIMUM_RATIO){
          if (ratioOldersecondHighest > MINIMUM_RATIO |
            ratioOldersecondHighest == Inf) {
            elderly_related_publications <- elderly_
            related_publications + 1
            # here we go to next one as this is already
            categorized
            next
          }

          # now check that all others are below threshold
          else if ((dim(result_disability_vision)[1] >
            MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
            disability_cognitive)[1] > MAXIMUM_OCCURRENCES_FOR_
            MANUAL) | (dim(result_disability_psychological)[1] >
            MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
            disability_mobility)[1] > MAXIMUM_OCCURRENCES_FOR_
            MANUAL) | (dim(result_disability_hearing)[1] >
            MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
            disability_general)[1] > MAXIMUM_OCCURRENCES_FOR_
            MANUAL)) {
            message(" Error for
            paper, manual adjustment needed: ")
            message(" ", files[i])
            # message(" ", metadata$title))
            number_of_problems <- number_of_problems + 1
          } else {
            # otherwise add one to the year and the
            correct column
            elderly_related_publications <- elderly_
            related_publications + 1

```

```

        next
      } else if (dim(result_disability_general)[1] >
        MINIMUM_OCCURRENCES_FOR_ACCEPTANCE) {

        # compute all ratios
        # ratioGeneralHearing <- dim(result_disability_
        general)[1] / dim(result_disability_hearing)[1]
        # ratioGeneralCognitive <- dim(result_
        disability_general)[1] / dim(result_disability_
        cognitive)[1]
        # ratioGeneralPsychological <- dim(result_
        disability_general)[1] / dim(result_disability_
        psychological)[1]
        # ratioGeneralMobility <- dim(result_disability_
        general)[1] / dim(result_disability_mobility)[1]
        # ratioGeneralOlder <- dim(result_disability_
        general)[1] / dim(result_disability_older)[1]
        # ratioGeneralVision <- dim(result_disability_
        general)[1] / dim(result_disability_vision)[1]

        secondHighest <- order(dim(result_disability_
        hearing)[1], dim(result_disability_cognitive)[1],
        dim(result_disability_psychological)[1], dim(result_
        disability_mobility)[1], dim(result_disability_older
        )[1], dim(result_disability_vision)[1])
        ratioGeneralsecondHighest <- dim(result_
        disability_general)[1] / secondHighest

        # if(ratioGeneralHearing > MINIMUM_RATIO &
        ratioGeneralCognitive > MINIMUM_RATIO &
        ratioGeneralPsychological > MINIMUM_RATIO &
        ratioGeneralMobility > MINIMUM_RATIO &
        ratioGeneralOlder > MINIMUM_RATIO &
        ratioGeneralVision > MINIMUM_RATIO){
          if (ratioGeneralsecondHighest > MINIMUM_RATIO |
            ratioGeneralsecondHighest == Inf) {
            general_related_publications <- general_
            related_publications + 1
            # here we go to next one as this is already
            categorized
            next
          }

          # now check that all others are below threshold
          else if ((dim(result_disability_vision)[1] >
            MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
            disability_cognitive)[1] > MAXIMUM_OCCURRENCES_FOR_
            MANUAL) | (dim(result_disability_psychological)[1] >
            MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
            disability_mobility)[1] > MAXIMUM_OCCURRENCES_FOR_
            MANUAL) | (dim(result_disability_hearing)[1] >
            MAXIMUM_OCCURRENCES_FOR_MANUAL) | (dim(result_
            disability_older)[1] > MAXIMUM_OCCURRENCES_FOR_
            MANUAL)) {
            message(" Error for
            paper, manual adjustment needed: ")
            message(" ", files[i])
            # message(" ", metadata$title))
            number_of_problems <- number_of_problems + 1
          } else {
            # otherwise add one to the year and the
            correct column
            general_related_publications <- general_
            related_publications + 1
            next
          }
        }
      }

      # if we come to this line, then this paper could
      not be categorized!
      number_of_not_categorized <- number_of_not_
      categorized + 1

```

```

message("Could not categorize: ", substr(files[i],
1, nchar(files[i]) - 4))

# we want to know the number of keywords for every
non-categorized work
if (TRUE) {
  # No rows
  # This data frame has columns but no observations
  .

  # [1] = rows
  # [2] = columns
  if (dim(result_disability_vision)[1] > 0) {
    # a term was found!
    message("Vision disability term was found.
Number:", dim(result_disability_vision)[1])
    # message(head(result_disability_vision))
  }

  if (dim(result_disability_hearing)[1] > 0) {
    # a term was found!
    message("Hearing disability term was found.
Number:", dim(result_disability_hearing)[1])
    # message(head(result_disability_hearing))
  }

  if (dim(result_disability_cognitive)[1] > 0) {
    # a term was found!
    message("Cognitive disability term was found.
Number:", dim(result_disability_cognitive)[1])
    # message(head(result_disability_cognitive))
  }

  if (dim(result_disability_psychological)[1] > 0)
  {
    # a term was found!
    message("Psychological disability term was
found. Number:", dim(result_disability_psychological
)[1])
    # message(head(result_disability_psychological
))
  }

  if (dim(result_disability_mobility)[1] > 0) {
    # a term was found!
    message("Mobility disability term was found.
Number:", dim(result_disability_mobility)[1])
    # message(head(result_disability_mobility))
  }

  if (dim(result_disability_older)[1] > 0) {
    # a term was found!
    message("Elderly disability term was found.
Number:", dim(result_disability_older)[1])
    # message(head(result_disability_older))
  }

  if (dim(result_disability_general)[1] > 0) {
    # a term was found in TITLE!
    message("General disability term was found.
Number:", dim(result_disability_general)[1])
    # message(head(result_disability_general))
  }
}

# create row for all found keywords
df_one_year_uncategorized <- data.frame(
  Conference = substring(conference_directories[d],
3),
  Vision = dim(result_disability_vision)[1],
  Hearing = dim(result_disability_hearing)[1],
  Cognitive = dim(result_disability_cognitive)[1],
  Psychological = dim(result_disability_
psychological)[1],
  Mobility = dim(result_disability_mobility)[1],
  General = dim(result_disability_general)[1],
  Elderly = dim(result_disability_older)[1],

  stringsAsFactors = FALSE
)

# string these together
df_occurrences_un_categorized <- rbind(df_
occurrences_un_categorized, df_one_year_
uncategorized)

#
if (accessibility_related_occurrences < 20) {
  unrelated_work <- unrelated_work + 1
} else {
  potentially_related_work <- potentially_related_
work + 1
}

# message("-----")
}

# now put all the variables in the right column
# as we start in 2009
row_for_df <- year - 2008

message("Making row for year: ", year)

# adjust YEAR as defined by the name of the folder
# [row][column]
df_conference[row_for_df, 1] <- as.character(year)
df_conference[row_for_df, 2] <- as.integer(vision_
related_publications)
df_conference[row_for_df, 3] <- as.integer(hearing_
related_publications)
df_conference[row_for_df, 4] <- as.integer(cognitive_
related_publications)
df_conference[row_for_df, 5] <- as.integer(
psychological_related_publications)
df_conference[row_for_df, 6] <- as.integer(mobility_
related_publications)
df_conference[row_for_df, 7] <- as.integer(general_
related_publications)
df_conference[row_for_df, 8] <- as.integer(elderly_
related_publications)

# go up a directory to directory with the years
setwd("../")
}

df_conference$Overall <- rowSums(df_conference[, c(2:8)
])

# column_sums
df_conference[row_for_df + 1, 1] <- as.character("
Combined")
df_conference[row_for_df + 1, c(2:9)] <- colSums(df_
conference[, c(2:9)], na.rm = TRUE)
message("Making formattable table")

# now make a nice coloured table from it
format <- formattable(df_conference, list(area(col =
2:8) ~ color_tile("grey", "green"))))
export_formattable(format, paste0(substring(conference_
directories[d], 3), "_table.png"))

# package DT needed
datatable_conference <- as.data.table(format)
DT::saveWidget(datatable_conference, paste0(substring(
conference_directories[d], 3), "_table.html"))

# go up a directory
setwd("../")
}

message("Analysis complete")
# 4 = "complete"
beep(sound = 4, expr = NULL)

```

```

830 message("In total, this many problems occurred: ", number
      _of_problems)
831 message("In total, this many papers could not be
      categorized: ", number_of_not_categorized)
832
833 message("In total, this many papers are probably
      unrelated: ", unrelated_work)
834 message("In total, this many papers are maybe related: ",
      potentially_related_work)
835
836 message("In total, this many papers were categorized by
      title: ", categorized_by_title)
837 message("In total, this many papers were categorized by
      keywords: ", categorized_by_keywords)
838
839
840 message(warnings())
841 sink()
842
843 df2 <- as.data.frame(df_occurrences)
844
845 df2$secondMax <- NULL
846 df2$thirdMax <- NULL
847 df2$fourthMax <- NULL
848 df2$fifthMax <- NULL
849 df2$sixthMax <- NULL
850
851 df2$max <- NULL
852 df2$min <- NULL
853
854 # now find the max and second most ... per paper
855 # this was used to assess the criteria for the
856 # categorization
857 # 1 indexes rows
858 df2$max <- apply(df2[, c(2:8)], 1, max)
859 df2$min <- apply(df2[, c(2:8)], 1, min)
860 df2$secondMax <- apply(df2[, c(2:8)], 1, function(x) x[
      maxn(2)(x)])
861 df2$thirdMax <- apply(df2[, c(2:8)], 1, function(x) x[
      maxn(3)(x)])
862 df2$fourthMax <- apply(df2[, c(2:8)], 1, function(x) x[
      maxn(4)(x)])
863 df2$fifthMax <- apply(df2[, c(2:8)], 1, function(x) x[
      maxn(5)(x)])
864 df2$sixthMax <- apply(df2[, c(2:8)], 1, function(x) x[
      maxn(6)(x)])
865
866 df2$Overall <- rowSums(df2[, c(2:8)])
867
868 df2$RatioMaxSecondMax <- df2$max / df2$secondMax
869
870 mean(df2$max)
871 sd(df2$max)
872 quantile(df2$max, 0.25)
873 quantile(df2$max, 0.1)
874
875 ggplot(df2, aes(x = max)) +
876   geom_histogram(aes(y = ..density..), # the histogram
877     will display "density" on its y-axis
878     binwidth = .5, colour = "grey", fill = "white"
879   ) +
880   geom_density(alpha = .2, fill = "#FF6655") +
881   geom_vline(aes(xintercept = mean(max, na.rm = T)),
882     colour = "red", linetype = "longdash", size = .8
883   ) +
884   geom_vline(aes(xintercept = quantile(max, 0.1)),
885     colour = "blue", linetype = "longdash", size = .5
886   ) +
887   geom_vline(aes(xintercept = quantile(max, 0.25)),
888     colour = "green", linetype = "longdash", size = .5
889   ) +
890   geom_vline(aes(xintercept = quantile(max, 0.75)),
891     colour = "green", linetype = "longdash", size = .5
892   ) +
893   scale_x_continuous(limits = c(00, 100))
894
895 ggsave("PATH_TO_SAVE.pdf", width = pdfwidth, height =
      pdfheight + 2, device = cairo_pdf)
896
897
898 mean(df2$secondMax)
899 sd(df2$secondMax)
900 quantile(df2$secondMax, 0.25)
901 quantile(df2$secondMax, 0.75)
902
903 ggplot(df2, aes(x = secondMax)) +
904   geom_histogram(aes(y = ..density..), # the histogram
905     will display "density" on its y-axis
906     binwidth = .5, colour = "grey", fill = "white"
907   ) +
908   geom_density(alpha = .2, fill = "#FF6655") +
909   geom_vline(aes(xintercept = mean(secondMax, na.rm = T)),
910     colour = "red", linetype = "longdash", size = .8
911   ) +
912   geom_vline(aes(xintercept = quantile(secondMax, 0.25)),
913     colour = "green", linetype = "longdash", size = .5
914   ) +
915   geom_vline(aes(xintercept = quantile(secondMax, 0.75)),
916     colour = "blue", linetype = "longdash", size = .5
917   ) +
918   ggtitle("Second most common occurrences") +
919   scale_x_continuous(limits = c(00, 30))
920
921 ggsave("PATH_TO_SAVE.pdf", width = pdfwidth, height =
      pdfheight + 2, device = cairo_pdf)
922
923 mean(df2$thirdMax)
924 mean(df2$fourthMax)
925
926 mean(df2$Overall)
927 sd(df2$Overall)
928 quantile(df2$Overall, 0.25)
929 quantile(df2$Overall, 0.05)
930
931
932 ggplot(df2, aes(x = Overall)) +
933   geom_histogram(aes(y = ..density..), # the histogram
934     will display "density" on its y-axis
935     binwidth = .5, colour = "grey", fill = "white"
936   ) +
937   geom_density(alpha = .2, fill = "#FF6655") +
938   geom_vline(aes(xintercept = mean(Overall, na.rm = T)),
939     colour = "red", linetype = "longdash", size = .8
940   ) +
941   geom_vline(aes(xintercept = quantile(Overall, 0.25)),
942     colour = "green", linetype = "longdash", size = .5
943   ) +
944   geom_vline(aes(xintercept = quantile(Overall, 0.75)),
945     colour = "green", linetype = "longdash", size = .5
946   ) +
947   geom_vline(aes(xintercept = quantile(Overall, 0.05)),
948     colour = "blue", linetype = "longdash", size = .5
949   ) +
950   ggtitle("Overall occurrences") +
951   scale_x_continuous(limits = c(00, 150))
952
953 ggsave("PATH_TO_SAVE.pdf", width = pdfwidth, height =
      pdfheight + 2, device = cairo_pdf)
954
955
956
957
958
959
960
961
962 ### only look at accessibility-related venues:

```

```

964 accessDf <- subset(df2, df2$Conference == "W4A" | df2$
    Conference == "ASSETS" | df2$Conference == "TACCESS"
    )
966
968 mean(accessDf$max)
970 sd(accessDf$max)
972 quantile(accessDf$max, 0.25)
974 quantile(accessDf$max, 0.1)
976
978 mean(accessDf$secondMax)
980 sd(accessDf$secondMax)
982 quantile(accessDf$secondMax, 0.25)
984 quantile(accessDf$secondMax, 0.50)
986 quantile(accessDf$secondMax, 0.75)
988
990 mean(accessDf$Overall)
992 sd(accessDf$Overall)
994 quantile(accessDf$Overall, 0.25)
996 quantile(accessDf$Overall, 0.05)
1000
1002
1004 ggplot(accessDf, aes(x = Overall)) +
    geom_histogram(aes(y = ..density..), # the histogram
        will display "density" on its y-axis
        binwidth = .5, colour = "grey", fill = "white"
    ) +
    geom_density(alpha = .2, fill = "#FF6655") +
    geom_vline(aes(xintercept = mean(max, na.rm = T)),
        colour = "red", linetype = "longdash", size = .8
    ) +
    geom_vline(aes(xintercept = quantile(max, 0.1)),
        colour = "blue", linetype = "longdash", size = .5
    ) +
    geom_vline(aes(xintercept = quantile(max, 0.25)),
        colour = "green", linetype = "longdash", size = .5
    ) +
    geom_vline(aes(xintercept = quantile(max, 0.75)),
        colour = "green", linetype = "longdash", size = .5
    )
1006
1008 ggsave("PATH_TO_SAVE.pdf", width = pdfwidth, height =
    pdfheight + 2, device = cairo_pdf)
1010
1012 ggplot(accessDf, aes(x = max)) +
    geom_histogram(aes(y = ..density..), # the histogram
        will display "density" on its y-axis
        binwidth = .5, colour = "grey", fill = "white"
    ) +
    geom_density(alpha = .2, fill = "#FF6655") +
    geom_vline(aes(xintercept = mean(max, na.rm = T)),
        colour = "red", linetype = "longdash", size = .8
    ) +
    geom_vline(aes(xintercept = quantile(max, 0.1)),
        colour = "blue", linetype = "longdash", size = .5
    ) +
    geom_vline(aes(xintercept = quantile(max, 0.25)),
        colour = "green", linetype = "longdash", size = .5
    ) +
    geom_vline(aes(xintercept = quantile(max, 0.75)),
        colour = "green", linetype = "longdash", size = .5
    )
1024
1026 ggsave("PATH_TO_SAVE.pdf", width = pdfwidth, height =
    pdfheight + 2, device = cairo_pdf)
1028
1030
1032 ggplot(accessDf, aes(x = secondMax)) +
    geom_histogram(aes(y = ..density..), # the histogram
        will display "density" on its y-axis
        binwidth = .5, colour = "grey", fill = "white"
    ) +

```

```

1036 geom_density(alpha = .2, fill = "#FF6655") +
    geom_vline(aes(xintercept = mean(max, na.rm = T)),
        colour = "red", linetype = "longdash", size = .8
    ) +
1038
1040 geom_vline(aes(xintercept = quantile(max, 0.1)),
        colour = "blue", linetype = "longdash", size = .5
    ) +
1042
1044 geom_vline(aes(xintercept = quantile(max, 0.25)),
        colour = "green", linetype = "longdash", size = .5
    ) +
1046
1048 geom_vline(aes(xintercept = quantile(max, 0.75)),
        colour = "green", linetype = "longdash", size = .5
    )
1050
1052 ggsave("PATH_TO_SAVE.pdf", width = pdfwidth, height =
    pdfheight + 2, device = cairo_pdf)
1054
1056 ### only for those that are not categorizable
1058
1060 df_uncategorized <- as.data.frame(df_occurrences_un_
    categorized)
1062
1064 df_uncategorized$secondMax <- NULL
1066 df_uncategorized$thirdMax <- NULL
1068 df_uncategorized$fourthMax <- NULL
1070 df_uncategorized$fifthMax <- NULL
1072 df_uncategorized$sixthMax <- NULL
1074
1076 df_uncategorized$max <- NULL
1078 df_uncategorized$min <- NULL
1080
1082 # now find the max and second most ... per paper
1084 # this was used to assess the criteria for the
    categorization
1086 # 1 indexes rows
1088 df_uncategorized$max <- apply(df_uncategorized[, c(2:8)],
    1, max)
1090 df_uncategorized$min <- apply(df_uncategorized[, c(2:8)],
    1, min)
1092 df_uncategorized$secondMax <- apply(df_uncategorized[, c
    (2:8)], 1, function(x) x[max(2)(x)])
1094 df_uncategorized$thirdMax <- apply(df_uncategorized[, c
    (2:8)], 1, function(x) x[max(3)(x)])
1096 df_uncategorized$fourthMax <- apply(df_uncategorized[, c
    (2:8)], 1, function(x) x[max(4)(x)])
1098 df_uncategorized$fifthMax <- apply(df_uncategorized[, c
    (2:8)], 1, function(x) x[max(5)(x)])
1100 df_uncategorized$sixthMax <- apply(df_uncategorized[, c
    (2:8)], 1, function(x) x[max(6)(x)])
1102
1104 df_uncategorized$Overall <- rowSums(df_uncategorized[, c
    (2:8)])
1106
1108 df_uncategorized$RatioMaxSecondMax <- df_uncategorized$
    max / df_uncategorized$secondMax
1110
1112 # get number of uncategorized paper
1114 nrow(df_uncategorized)
1116
1118 mean(df_uncategorized$max)
1120 sd(df_uncategorized$max)
1122 quantile(df_uncategorized$max, 0.25)
1124 quantile(df_uncategorized$max, 0.1)
1126
1128
1130 ggplot(df_uncategorized, aes(x = max)) +
    geom_histogram(aes(y = ..density..), # the histogram
        will display "density" on its y-axis
        binwidth = .5, colour = "grey", fill = "white"
    ) +

```

Accessibility-Related Publication Distribution in HCI Based on a Meta-Analysis

CHI '22 Extended Abstracts, April 29-May 5, 2022, New Orleans, LA, USA

```

1100 geom_density(alpha = .2, fill = "#FF6655") +
1101 geom_vline(aes(xintercept = mean(max, na.rm = T)),
1102   colour = "red", linetype = "longdash", size = .8
1103 ) +
1104 geom_vline(aes(xintercept = quantile(max, 0.1)),
1105   colour = "blue", linetype = "longdash", size = .5
1106 ) +
1107 geom_vline(aes(xintercept = quantile(max, 0.25)),
1108   colour = "green", linetype = "longdash", size = .5
1109 ) +
1110 geom_vline(aes(xintercept = quantile(max, 0.75)),
1111   colour = "green", linetype = "longdash", size = .5
1112 )
1113 ggsave("PATH_TO_SAVE.pdf", width = pdfwidth, height =
1114   pdfheight + 2, device = cairo_pdf)
1115
1116
1117 mean(df_uncategorized$secondMax)
1118 sd(df_uncategorized$secondMax)
1119 quantile(df_uncategorized$secondMax, 0.25)
1120 quantile(df_uncategorized$secondMax, 0.75)
1121
1122 ggplot(df_uncategorized, aes(x = secondMax)) +
1123   geom_histogram(aes(y = ..density..), # the histogram
1124     will display "density" on its y-axis
1125     binwidth = .5, colour = "grey", fill = "white"
1126   ) +
1127   geom_density(alpha = .2, fill = "#FF6655") +
1128   geom_vline(aes(xintercept = mean(secondMax, na.rm = T))
1129     , colour = "red", linetype = "longdash", size = .8
1130   ) +
1131   geom_vline(aes(xintercept = quantile(secondMax, 0.25)),
1132     colour = "green", linetype = "longdash", size = .5
1133   ) +
1134   geom_vline(aes(xintercept = quantile(secondMax, 0.75)),
1135     colour = "blue", linetype = "longdash", size = .5
1136   ) +
1137   ggtitle("Second most common occurrences")
1138 ggsave("PATH_TO_SAVE.pdf", width = pdfwidth, height =
1139   pdfheight + 2, device = cairo_pdf)
1140
1141
1142
1143
1144
1145
1146 ggplot(df_uncategorized, aes(x = Overall)) +
1147   geom_histogram(aes(y = ..density..), # the histogram
1148     will display "density" on its y-axis
1149     binwidth = .5, colour = "grey", fill = "white"
1150   ) +
1151   geom_density(alpha = .2, fill = "#FF6655") +
1152   geom_vline(aes(xintercept = mean(max, na.rm = T)),
1153     colour = "red", linetype = "longdash", size = .8
1154   ) +
1155   geom_vline(aes(xintercept = quantile(max, 0.1)),
1156     colour = "blue", linetype = "longdash", size = .5
1157   ) +
1158   geom_vline(aes(xintercept = quantile(max, 0.25)),
1159     colour = "green", linetype = "longdash", size = .5
1160   ) +
1161   geom_vline(aes(xintercept = quantile(max, 0.75)),
1162     colour = "green", linetype = "longdash", size = .5
1163   )
1164 ggsave("PATH_TO_SAVE.pdf", width = pdfwidth, height =
1165   pdfheight + 2, device = cairo_pdf)
1166
1167
1168
1169
1170

```

```

1172
1173 mean(subset(df_uncategorized$max, df_uncategorized$
1174   Conference == "W4A"))
1175 sd(subset(df_uncategorized$max, df_uncategorized$
1176   Conference == "W4A"))
1177
1178 mean(subset(df_uncategorized$secondMax, df_uncategorized$
1179   Conference == "W4A"))
1180 sd(subset(df_uncategorized$secondMax, df_uncategorized$
1181   Conference == "W4A"))
1182
1183 quantile(subset(df_uncategorized$max, df_uncategorized$
1184   Conference == "W4A"), 0.99)
1185
1186
1187 w4a <- subset(df_uncategorized, df_uncategorized$
1188   Conference == "W4A")
1189 nrow(w4a)
1190
1191 taccess <- subset(df_uncategorized, df_uncategorized$
1192   Conference == "TACCESS ")
1193 nrow(taccess)
1194
1195 assets <- subset(df_uncategorized, df_uncategorized$
1196   Conference == "ASSETS ")
1197 nrow(assets)
1198
1199 uist <- subset(df_uncategorized, df_uncategorized$
1200   Conference == "UIST")
1201 nrow(uist)
1202
1203 chi <- subset(df_uncategorized, df_uncategorized$
1204   Conference == "CHI")
1205 nrow(chi)
1206
1207 autoui <- subset(df_uncategorized, df_uncategorized$
1208   Conference == "AutoUI")
1209 nrow(autoui)
1210
1211
1212 potentially_related_work

```

paper_evaluation_anonymized.R